Lecture Scribe Joint Variables

L15_L17.pdf
PDF

Generate a lecture scribe intended to serve as accurate reference material for exam preparation.
Use only the provided lecture slides and relevant textbook content as the authoritative context.

The lecture scribe must:

Faithfully reconstruct what was taught in the lecture without introducing any new information.

Include all definitions, notation, assumptions, theorem or result statements, proofs or derivations, and worked examples only if they appear in the provided context.

Present proofs, derivations, and example solutions step by step exactly as shown in the lecture material.

Maintain a clear logical flow that allows a student to reliably revise the lecture using only this document.

Follow a formal academic tone suitable for closed-notes exam preparation.

Be clearly structured with appropriate section headings and organized progression of ideas.

The lecture scribe must not:

Add intuition, explanations, examples, or results that are not explicitly present in the provided context.

Introduce creative rewriting, simplifications, summaries, or external knowledge.

Include demonstrations, sample scribes, or reasoning instructions.

Ensure strict correctness, completeness, and full alignment with the lecture and textbook materials.

Lecture Scribe: Joint Random Variables and Joint Distributions

Course: CSE400 – Fundamentals of Probability in Computing
Lectures: L15–L18
Instructor: Dhaval Patel, PhD
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1. Motivation: Why Study Joint Random Variables

In science and real-life applications, we are often interested in **two (or more) random variables simultaneously**.

Examples include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

Illustrative Applications

- **Smart Transportation:**
  $X =$ Vehicle Speed, $Y =$ Traffic Density
  These variables may be correlated.
- **Wireless Networks:**
  $X =$ Signal-to-Noise Ratio (SNR), $Y =$ Data Throughput
  Throughput depends on SNR.
- **Weather Prediction:**
  $X =$ Rainfall, $Y =$ Temperature
- **Drone Delivery:**
  $X =$ Wind Speed, $Y =$ Delivery Time

- **Dice Example:**
  Joint behavior of two dice requires joint PMF.

Key Statement

- Joint distributions allow computation of probabilities involving multiple variables and understanding their **relationship**.
- If variables are not independent, **covariance and correlation** are used to measure dependence.

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2. Discrete Case – Joint PMF

Setup

Let X and Y be discrete random variables:

$$X = \{x_1, x_2, \ldots, x_n\}, \quad Y = \{y_1, y_2, \ldots, y_m\}$$

The ordered pair (X, Y) takes values in the product set:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This represents the probability of the joint outcome occurring simultaneously.

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3. Joint PMF Table Representation

The joint PMF can be organized in a table with rows for X and columns for Y.

Key Properties

$$0 \leq p(x_i, y_j) \leq 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

4. Example: Rolling Two Dice

Experiment

Roll two fair dice.

Let:

- $X =$ value on the first die
- $T =$ total (sum) of both dice

Each outcome (i, j) has probability:

$$\frac{1}{36}$$

A joint probability table of (X, T) is constructed (shown on page 14).

Observations

- Entries are either 0 or $\frac{1}{36}$
- X and T are **not independent**

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5. Continuous Case – Joint PDF

Analogy with Discrete Case

- Discrete values → Continuous intervals
- Joint PMF → Joint PDF
- Sums → Double integrals

If:

$$X \in [a, b], \quad Y \in [c, d]$$

then:

$$(X, Y) \in [a, b] \times [c, d]$$

Definition (Joint PDF)

A function $f(x, y)$ is called the joint probability density function if:

$$P((X, Y) \text{ lies in a small rectangle of area } dx\,dy) \approx f(x, y)\,dx\,dy$$

Key Properties

$$f(x, y) \geq 0$$

$$\int_c^d \int_a^b f(x, y)\,dx\,dy = 1$$

Note:
The joint PDF may take values greater than 1 since it is a density, not a probability.

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6. Worked Example (Continuous Case)

Given

$$f(x, y) = 4xy, \quad 0 \leq x \leq 1,\ 0 \leq y \leq 1$$

Tasks:

1. Show that $f(x, y)$ is a valid joint PDF
2. Visualize event:

$$A = \{X < 0.5,\ Y > 0.5\}$$

3. Compute $P(A)$

Joint domain: unit square $[0, 1]^2$.

Step 1: Check Validity (Normalization)

$$\int_0^1 \int_0^1 4xy\,dx\,dy = \int_0^1 \left(\int_0^1 4xy\,dx \right) dy = \int_0^1 2y\,dy = [y^2]_0^1 = 1$$

Hence $f(x, y) \geq 0$ and total probability is 1.
Therefore, it is a valid joint PDF.

Step 2: Compute $P(A)$

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

$$= \int_0^{0.5} \left[2xy^2\right]_{0.5}^{1} dx = \int_0^{0.5} 2x(1 - 0.25) \, dx$$

$$= \int_0^{0.5} \frac{3}{2} x \, dx = \frac{3}{16}$$

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7. Joint Cumulative Distribution Function (Joint CDF)

Notation

$$X \leq x, \; Y \leq y \quad \text{means} \quad \{X \leq x \text{ and } Y \leq y\}$$

Definition

$$F(x, y) = P(X \leq x, \; Y \leq y)$$

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8. Joint CDF – Continuous Case

If X, Y have joint density $f(x, y)$ over $[a, b] \times [c, d]$, then:

$$F(x, y) = \int_c^y \int_a^x f(u, v) \, du \, dv$$

Recovering Joint PDF

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

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9. Joint CDF – Discrete Case

If X, Y are discrete with joint PMF $p(x_i, y_j)$, then:

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

Interpretation

- Sum probabilities in the table
- Include rows with $x_i \leq x$
- Include columns with $y_j \leq y$

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10. Properties of the Joint CDF

The joint CDF $F(x, y)$ satisfies:

1. **Non-decreasing:**

$$F(x, y) \text{ increases or remains constant as } x \text{ or } y \text{ increase}$$

2. **Lower-left limit:**

$$F(x, y) = 0 \text{ at } (-\infty, -\infty)$$

3. **Upper-right limit:**

$$F(x, y) = 1 \text{ at } (\infty, \infty)$$

End of Lecture